

STRATEGIC ELECTRONICS Set For Revamp



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Large-scale modernisation of the Indian defence forces is on the anvil, making strategic electronics (mainly comprising aerospace and defence electronics) a vibrant industry. India needs to modernise its aerospace and defence capital equipment base by addressing obsolescence as well as building additional capabilities through new capital acquisition. The new Defence Procurement Policy (DPP) and offset policies announced by the Indian government are a big step in that direction. The government has also made budgetary allocations for paramilitary and state police force requirements. All these developments have made India quite attractive for aerospace and defence electronics manufacturers.

Market opportunities

As many as 80 per cent (Fig. 1) of the survey participants feel that the industry will see a slow but steady growth in the coming decade. There will be an exponential growth in both standalone systems (as part of platforms) as well as other systems at sub-system level.

According to a report by consultancy firm Roland Berger, in partnership with the National Association of Software & Services Companies (NASSCOM) and India Electronics and Semiconductor Association (IESA), India's aerospace and defence industry is expected to consume electronics worth \$70-72 billion

in the next decade, as the country rapidly modernises its military by embracing new technologies. Electronics acquired as part of platforms (at Tier 1 and Tier 2 levels) will be worth \$53-54 billion. Demand for electronics worth \$17-18 billion is expected from projects that are traditionally called system-of-systems (non-platform based) applications.

Another growth opportunity is the potential to integrate into global value chains of original equipment manufacturers (initially leveraging the offset route and subsequently based on product quality, service delivery advantages and cost arbitrage).

Market demand

In spite of huge opportunities, there is a demand-supply gap for indigenous components, products and solutions, most of which are still imported. Indian companies are not able to cater to requirements that involve low volumes, high technology and high investments.

Areas that require electronic products and solutions in large volumes include:

1. Electronics manufacturing services, 'build to print' (which, in the context of defence production, includes firmware updates, test-jig development, manufacturing software development, component procurement, systems integration, software upgrades/enhancement, verification and validation, system deployment, hardware qualification, environmental stress screening, maintenance support, etc), line replaceable units for Indian programmes like Light Combat Aircraft (LCA) and Light Combat Helicopter (LCH), drones, Kamov helicopters, etc
2. Rugged displays, PDAs and laptops for battlefield management system project
3. Electronic components, products and solutions used in vehicles for projects like the Futuristic Infantry Combat Vehicle and self-propelled guns

Most of the recent upgrades in military systems have a huge scope for electronics, as in the case of Jaguar, Ilyushin aircrafts, MIG 29 and T-90 tanks. In addition, new ventures

Methodology

This report is based on inputs from 24 senior professionals involved in the strategic electronics ecosystem. Responses were taken either directly or from their speeches at *Defcon* conference conducted by the Electronic Industries Association of India (ELCINA) in April this year. This sample group is a microcosmic representation of India's strategic electronics ecosystem.

Survey participants shared their insights on:

1. Growth of the Indian defence electronics industry
2. Demand-generating applications
3. Possibility of public-private partnership (PPP) or industry-academia partnership
4. Impact of policy amendments
5. Challenges

A trend analysis was done on the basis of inputs. Results of the analysis are presented here.